

Task 9: Report on Vuln-Bank

Prepared for: MuLearn Bootcamp

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Description: Clone and deploy the purposely vulnerable Bank web app, inspect it for issues, and produce a concise professional vulnerability report.

Repo URL: <https://github.com/Commando-X/vuln-bank.git>

Setup Command: `git clone https://github.com/Commando-X/vuln-bank.git && cd vuln-bank && docker-compose up --build -d`

This report focuses on the vulnerabilities identified by using the vuln-bank repo from GitHub. The server runs using the Docker service on our localhost.

Setup Process

By installing the vuln-bank into Docker using the setup command, we can access the local site after successfully setting up our machine. The initial interface allows creating a new account (e.g., username: "test", password: "test") with a default balance of \$1000.

Observations

While testing, I found that the app does not handle user inputs properly and allows unsafe actions. These mistakes can let attackers take control or change account data.

Main Vulnerabilities Found

1. Login Bypass (SQL Injection)

- **Problem:** The login box accepts special characters like ' OR 1=1 --, allowing anyone to log in without a real password.
- **Effect:** Attackers can log in as admin without knowing the password.

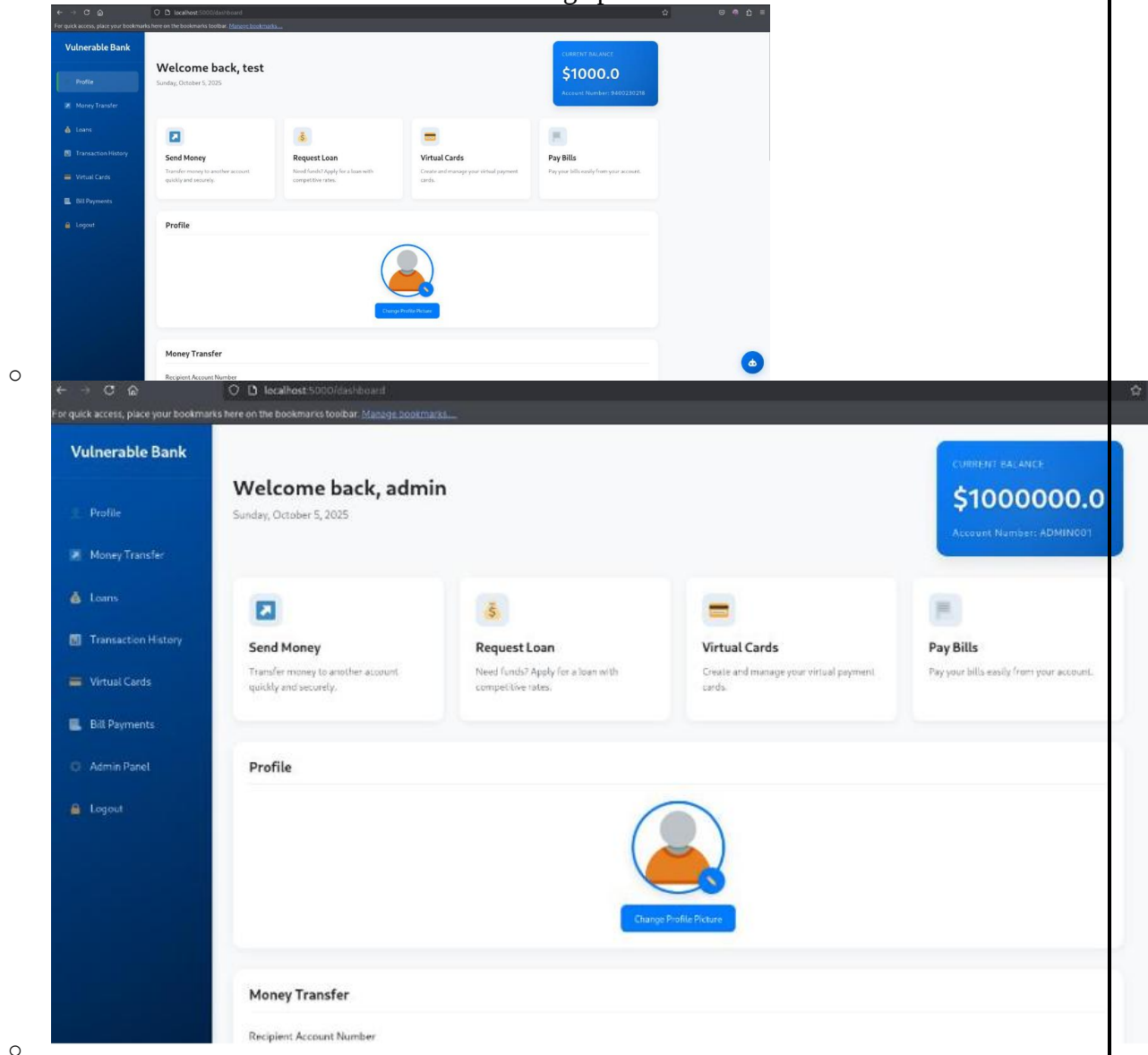


2. Account Editing from Browser

- **Problem:** When sending data to the server, users can change fields like "is_admin" or "balance" using browser tools.
- **Effect:** Attackers can make themselves admin or increase their account balance.
- **Screenshot:** [Image of Burp Suite showing account editing with modified "is_admin" and "balance"]

3. Admin Panel Not Protected Well

- **Problem:** Once inside the app, admin pages can be opened without strong security checks.
- **Effect:** Unauthorized users can see and change private data.



Impact

If this were a real bank, attackers could:

- Log in as admin without permission

- Change their money balance
- Approve or delete accounts
- View other users' information

Admin Control Panel


System Administrator

User Management

ID	Username	Account Number	Balance	Admin	Actions
1	admin	ADMIN001	\$1000000.00	True	Delete
2	test	9400230218	\$1000.00	False	Delete

Conclusion

The Vuln Bank app has several major security problems. It should never be used in real life without fixing them. This project helped me understand how insecure coding can make a web app vulnerable and how hackers can take advantage of it.

Vulnerable Bank

Welcome back, test
Sunday, October 5, 2025

CURRENT BALANCE
\$11000.0
Account Number: 9400230218


Send Money
Transfer money to another account quickly and securely.


Request Loan
Need funds? Apply for a loan with competitive rates.


Virtual Cards
Create and manage your virtual payment cards.


Pay Bills
Pay your bills easily from your account.

Profile


[Change Profile Picture](#)